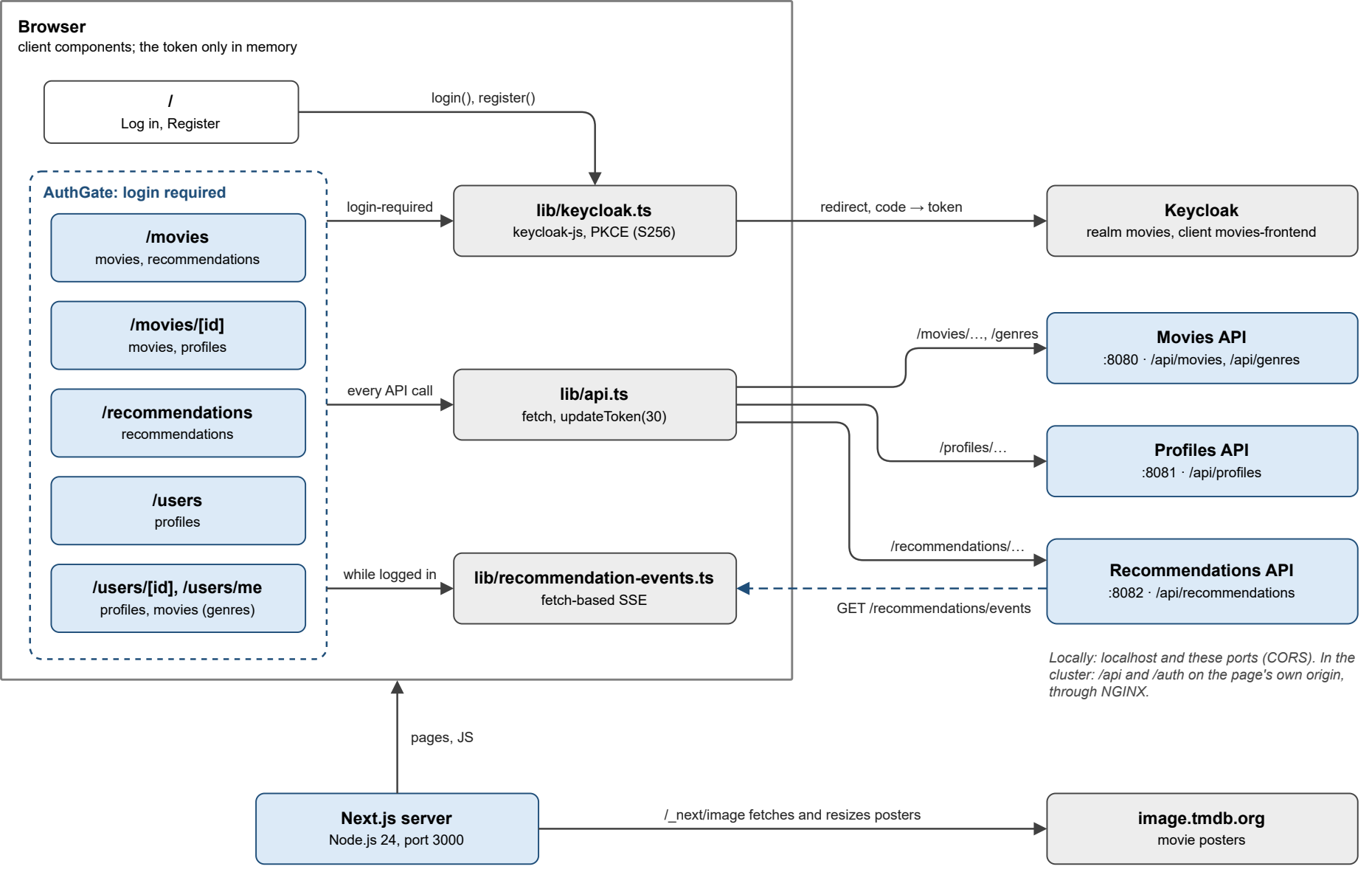

Browser
client components; the token only in memory
/
Log in, Register
login(), register()
AuthGate: login required
/movies
movies, recommendations
/movies/[id]
movies, profiles
/recommendations
recommendations
/users
profiles
/users/[id], /users/me
profiles, movies (genres)
login-required
lib/keycloak.ts
keycloak-js, PKCE (S256)
redirect, code → token
Keycloak
realm movies, client movies-frontend
every API call
lib/api.ts
fetch, updateToken(30)
/movies/…, /genres
Movies API
:8080 · /api/movies, /api/genres
/profiles/…
Profiles API
:8081 · /api/profiles
/recommendations/…
Recommendations API
:8082 · /api/recommendations
while logged in
lib/recommendation-events.ts
fetch-based SSE
GET /recommendations/events
Locally: localhost and these ports (CORS). In the cluster: /api and /auth on the page's own origin, through NGINX.
pages, JS
Next.js server
Node.js 24, port 3000
/_next/image fetches and resizes posters
image.tmdb.org
movie posters
page / service
shared module, platform
request
server-sent events
a page's second line: the APIs it calls